

AI/ML intern DAY 8

Research on AI-Based Virtual Try-On Models and Garment Fitting Technologies

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Introduction

The rapid advancement of Artificial Intelligence and Computer Vision has transformed the fashion industry by introducing Virtual Try-On (VTON) systems. These systems enable users to digitally wear garments without physically trying them on, creating a more interactive and personalized online shopping experience.

Modern Virtual Try-On systems combine multiple AI technologies including human detection, body segmentation, pose estimation, garment warping, and image synthesis to generate realistic outputs.

This research focuses on understanding the core technologies behind AI-based Virtual Try-On models and garment fitting systems.

Objectives

The objectives of this research are:

- Study AI-powered Virtual Try-On systems.
 - Understand garment fitting technologies.
 - Analyze body detection and pose estimation.
 - Explore garment warping methods.
 - Understand image synthesis techniques.
 - Perform a simple practical implementation.
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1. Virtual Try-On Technology

1.1 Overview

Virtual Try-On technology allows users to upload their image and visualize how clothing would appear on their body.

The system processes both the user image and garment image before generating a realistic fitting result.

Modern applications include:

- Online Fashion Stores
 - Smart Retail Platforms
 - Digital Dressing Rooms
 - Fashion Recommendation Systems
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1.2 Working Principle

A typical AI-based Virtual Try-On system follows these stages:

1. User Image Acquisition
 2. Human Body Detection
 3. Body Segmentation
 4. Pose Estimation
 5. Garment Processing
 6. Garment Warping
 7. Image Generation
 8. Final Virtual Try-On Output
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2. Human Body Detection

Human body detection is the first step in the Virtual Try-On process.

Its objective is to locate the human subject inside an image.

Popular approaches include:

MediaPipe Pose

- 33 body landmarks
- Lightweight architecture
- Real-time performance

OpenPose

- Multi-person support
 - High landmark accuracy
 - Suitable for research applications
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3. Body Segmentation

Body segmentation divides the image into different regions.

Common segmented parts include:

- Hair
- Face
- Upper Body
- Arms
- Legs
- Background

Segmentation helps AI systems understand where garments should be placed.

4. Pose Estimation

Pose estimation identifies the positions of important body joints.

Typical landmarks include:

- Left Shoulder
- Right Shoulder
- Elbows
- Wrists
- Hips
- Knees
- Ankles

Pose estimation improves garment alignment and body measurement estimation.

5. Garment Processing

The garment image must be prepared before fitting.

The processing stage includes:

- Background removal
- Image normalization
- Size adjustment
- Clothing masking

Proper garment preprocessing improves fitting quality.

6. Garment Warping

Garment warping adapts clothing to different body shapes.

Popular methods include:

Thin Plate Spline (TPS)

TPS uses control points to smoothly deform garments.

Advantages:

- Smooth transformations
 - Lightweight computation
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Geometric Matching Module (GMM)

GMM learns body-to-garment correspondences using deep learning.

Benefits:

- Better fitting accuracy
 - Improved garment adaptation
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Flow-Based Warping

Flow-based methods predict pixel movements between garment and body.

Advantages:

- Preserves texture
 - Handles complex poses
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7. Image Synthesis

Image synthesis combines the processed garment with the user image.

Objectives:

- Preserve garment texture.
- Maintain lighting consistency.
- Generate realistic folds.
- Blend garment edges smoothly.

Modern deep learning models significantly improve synthesis quality.

8. AI Fitting Systems

AI fitting systems estimate user body proportions.

Common measurements include:

- Shoulder Width
- Chest Width
- Waist Width
- Hip Width
- Arm Length

These measurements improve personalization and recommendation quality.

9. Practical Implementation

Basic Virtual Try-On Prototype

A simple OpenCV implementation was developed to simulate the basic Virtual Try-On process.

Python Code

```
import cv2

person = cv2.imread("person.jpg")
shirt = cv2.imread("shirt.png")

if person is None:
    print("person.jpg not found")
    exit()

if shirt is None:
    print("shirt.png not found")
    exit()

# Resize images
```

```
person = cv2.resize(person, (600, 800))
shirt = cv2.resize(shirt, (300, 250))

# Shirt position
x = 160
y = 150

h, w, _ = shirt.shape

# Overlay shirt
person[y:y+h, x:x+w] = shirt

# Save output
cv2.imwrite(
    "virtual_tryon_output.jpg",
    person
)

print("Virtual Try-On Generated Successfully")

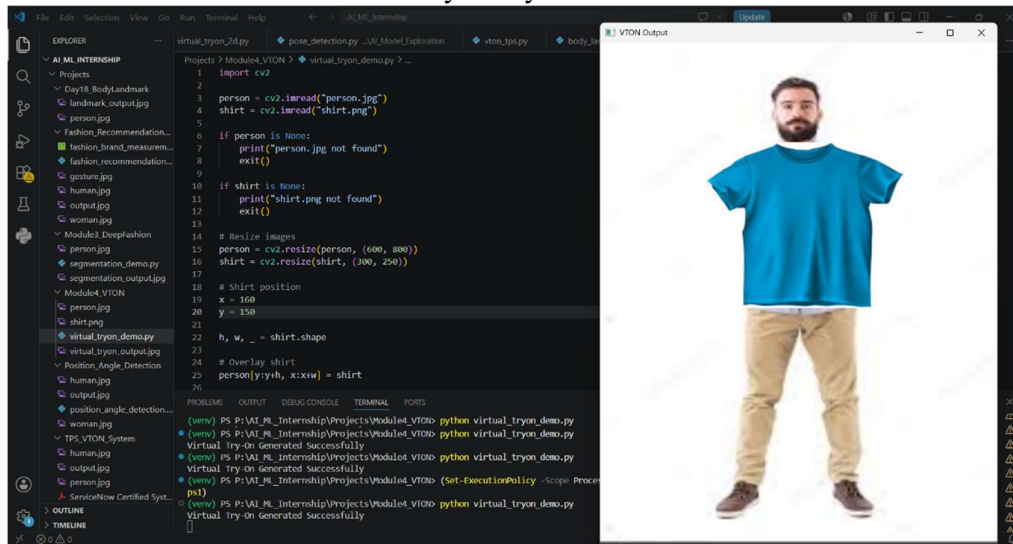
cv2.imshow(
    "VTON Output",
    person
)

cv2.waitKey(0)
cv2.destroyAllWindows()
```

Code Explanation

The implementation loads a user image and a garment image, resizes both inputs, and overlays the garment onto the upper body region.

Although this implementation does not use advanced AI-based warping algorithms, it demonstrates the basic concept of garment placement and image composition used in Virtual Try-On systems.



Output:



10. Applications

Virtual Try-On technology is widely used in:

Fashion E-commerce

Customers can preview garments before purchasing.

Smart Retail

Interactive shopping experiences improve engagement.

Digital Wardrobes

Users can organize and visualize their clothing collections.

AI Fashion Assistants

Recommendation systems provide personalized outfit suggestions.

11. Research Findings

The study indicates that successful Virtual Try-On systems require the integration of multiple AI modules including body detection, segmentation, pose estimation, garment warping, and image synthesis.

Among these components, garment warping and image synthesis play a major role in generating realistic outputs.

Advancements in deep learning continue to improve fitting accuracy and overall user experience.

12. Learning Outcomes

During this research, the following concepts were studied:

- Virtual Try-On Systems
 - Computer Vision
 - Human Detection
 - Body Segmentation
 - Pose Estimation
 - Garment Warping
 - Image Synthesis
 - AI Fashion Technologies
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13. Conclusion

This research explored the technologies behind AI-based Virtual Try-On systems and garment fitting methods. The study demonstrated how multiple computer vision modules work together to generate personalized clothing visualizations.

The findings highlight the growing importance of Artificial Intelligence in digital fashion and suggest that future developments in deep learning and computer vision will further enhance the realism and effectiveness of Virtual Try-On platforms.